JOINS OF GROTHENDIECK TOPOLOGIES IN THE LATTICE OF TWO-OUT-OF-THREE CLASSES: THE FINITE-ITERATION CRITERION

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Abstract. For a small category S, every Grothendieck topology J defines the class W_J of maps of presheaves inverted by sheafification, an element of the complete lattice $\mathrm{Same}(\mathrm{Presh}(S))$ of classes of morphisms closed under composition and two-out-of-three. The map $J \mapsto W_J$ preserves all meets; whether it preserves the join of two topologies was known to depend on the category. We describe the join $W_J \vee W_{J'}$ for arbitrary topologies J, J': it is the class of maps whose image sieves and equaliser sieves are reached by a bounded number of alternating applications of the closure operators j, j' of the two topologies, and every such map is a composite of maps of $W_J \cup W_{J'}$, so the join is the closure of the union under composition alone. Consequently $W_J \vee W_{J'} = W_{J \vee J'}$ if and only if $(jj')^k = j \vee j'$ for some k, the finite-iteration axiom for Lawvere–Tierney topologies, the same condition that governs joins of nuclei on a Heyting algebra and on the algebra of truth values of fuzzy sets. For topologies of ideal type, J_D and J_E for idempotent ideals D, E, the condition is the stabilisation of the chain $DE \supseteq DEDE \supseteq \cdots$, which contains the known positive answer for finite categories and the negative answer on the natural numbers; for ideals generated by two sets of objects A, B it decides when the join of "bijective at A" and "bijective at B" is "bijective at $A \cap B$"; on the natural numbers, the join of the classes of maps bijective at the even and at the odd points is the class of maps of towers that are bijective up to a uniform shift, a proper subclass of the isomorphisms of pro-sets.

1. Introduction

Let S be a small category and $\mathrm{Presh}(S)$ its category of presheaves. A Grothendieck topology J on S has a sheafification functor a_J, and the class W_J of maps inverted by a_J, the J-local isomorphisms, is a *sameness*: a class of morphisms containing the identities and closed under composition and two-out-of-three. Samenesses form a complete lattice $\mathrm{Same}(\mathrm{Presh}(S))$ under inclusion, meets being intersections and the join of two classes the

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least sameness containing both [5]. The assignment $J \mapsto W_J$ preserves all meets [7, Thm. 3.1], and the question studied in [7, 8] is whether it preserves binary joins: whether every map inverted by the sheafification of $J \vee J'$ is obtained from J-local and J'-local isomorphisms by finitely many compositions and cancellations. The answer was known to be yes for every finite category [8] and no for the poset $\mathbb{N}$ [7, Thm. 7.1]. This paper describes the join $W_J \vee W_{J'}$ for arbitrary topologies and gives the exact criterion for it to equal $W_{J\vee J'}$.

A topology J has a closure operator j on sieves: for a sieve S on c, $j(S)$ is the sieve of morphisms b into c with b^*S covering; S is covering iff $j(S)$ is the maximal sieve. A map $f\colon X \to Y$ of presheaves is in W_J iff its *image sieves* $\{b : y \cdot b \in \mathrm{im} f\}$, $y \in Y(c)$, and its *equaliser sieves* $\{b : x \cdot b = x' \cdot b\}$, $f(x) = f(x')$, are covering. Say a sieve is *k-reachable* for J, J' if $(jj')^k$ sends it to the maximal sieve, and let U_k be the class of maps all of whose image and equaliser sieves are k-reachable.

Theorem 1.1. *For any topologies J, J' on a small category S,*

$$W_J \vee W_{J'} = \bigcup_k U_k,$$

U_0 is the class of isomorphisms, and for $k \geq 1$ every map in U_k is a composite of $2k$ surjections followed by $2k$ injections lying alternately in $W_{J'}$ and W_J. In particular the join is the closure of $W_J \cup W_{J'}$ under composition.

Theorem 1.2. *$W_J \vee W_{J'} = W_{J\vee J'}$ if and only if $(jj')^k = j_{J\vee J'}$ on sieves for some k, i.e. $(jj')^k = j \vee j'$ as Lawvere–Tierney topologies on* $\mathrm{Presh}(S)$*; equivalently, every $(J \vee J')$-covering sieve is k-reachable for one k.*

The closure operator of $J \vee J'$ is the transfinite iterate of jj' (Lemma 2.2), so Theorem 1.2 says that the join of the two classes is the class of the join topology exactly when the transfinite iteration stops at a finite stage uniformly. This is the condition $(j_2 j_1)^N = j_1 \vee j_2$ that governs joins of nuclei on a Heyting algebra and their joins in the sameness lattice of fuzzy sets [9, 11], now for the internal frame of a presheaf topos.

For topologies of *ideal type* the criterion becomes a chain condition. A two-sided ideal D of S is a set of morphisms closed under composition with arbitrary morphisms on either side; it is *idempotent* if every member is a composite of two members. Every idempotent ideal D defines a topology J_D whose covering sieves on c are the sieves containing $D(-, c) = \{a\colon x \to c \text{ in } D\}$ [6, Thm. 5.1]; on a finite category every topology is of this form (Kelly–Lawvere [2], [4]), on an infinite one not necessarily. For ideals D, E write $DE = \{d \circ e : d \in D, e \in E\}$, again an ideal, and $(DE)^k = DE \cdots DE$ (k factors of each).

Corollary 1.3. *Let D, E be idempotent ideals of S. Then $W_{J_D \vee J_E} = W_{J_D} \vee W_{J_E}$ if and only if the chain $DE \supseteq (DE)^2 \supseteq \cdots$ stabilises, i.e. $(DE)^m = (DE)^{m+1}$ for some m. In that case $I' := (DE)^m$ is an idempotent ideal*

with $(ED)^{m+1} = I'$, $J_D \vee J_E = J_{I'}$, $W_{J_D} \vee W_{J_E} = W_{J_{I'}}$, *and every* $J_{I'}$*-local isomorphism is a composite of* $4m$ *maps of* $W_{J_D} \cup W_{J_E}$. *In every case* $W_{J_D} \vee W_{J_E}$ *is the class of maps locally bijective along some power* $(DE)^k$.

On a finite category the chain stabilises for every pair, which recovers [8] with a proof of a few lines in place of the separated quotients and plus towers used there; on $\mathbb{N}$ with the ideals of the even and of the odd points it is strictly decreasing, which recovers [7, Thm. 7.1]. The criterion becomes concrete for ideals generated by objects. For a set A of objects, the ideal D_A of morphisms factoring through A is idempotent, and $W_A = W_{J_{D_A}}$ is the class of maps bijective at every object of A, the class inverted by restriction to A, the sheafification for the essential subtopos $\mathrm{Presh}(A) \hookrightarrow \mathrm{Presh}(S)$. Proposition 6.1 decides when $W_A \vee W_B = W_{A \cap B}$: iff every morphism factoring through a long enough alternating zigzag of objects of B and A factors through an object of $A \cap B$, on a locally finite poset iff the strictly alternating chains avoiding $A \cap B$ have bounded size. On $\mathbb{N}$ with the even and the odd points the join is the class of maps of towers that are bijective up to a uniform shift, a proper subclass of the isomorphisms of pro-sets (Example 6.3).

The proofs are elementary. The inclusion $U_k \subseteq W_J \vee W_{J'}$ is a peeling argument: a surjection whose equaliser sieves are k-reachable is divided by the equivalence relation "equal under f and J'-locally equal", which is a J'-local isomorphism, and the induced map has equaliser sieves reachable in one step fewer; dually an injection is enlarged to the J'-closure of its image. The failure direction uses, for a covering sieve S that is not k-reachable, the fold map of two copies of a representable glued along S; the coproduct of these maps over all k is inverted by the join sheafification but lies in no U_k.

What is known. Samenesses and W_J are from [5]; that meets of topologies go to meets of samenesses, the reduction lemma and the $\mathbb{N}$ counterexample are from [7]; the finite case, with the alternating separated quotients, the plus tower and the retraction of representing bimodules, is [8]. Ideal topologies, the correspondence with idempotent ideals on finite categories and the local operators $R \mapsto \{a : a \cdot D \subseteq R\}$ are in [2, 4, 6]. Local isomorphisms as locally injective and locally surjective maps are standard [3, III.7]; joins of nuclei as transfinite iterates are in [1]. The first version of this paper (archived at DOI 10.5281/zenodo.23467281) proved Corollary 1.3 for ideal topologies only, with the stabilising direction by the route of [8].

What is new. Theorem 1.1, the description of the join for arbitrary topologies and the fact that no cancellations are needed; Theorem 1.2, the finite-iteration criterion, which decides the question for every pair of topologies on every small category, whereas [8] proves the positive answer under a finiteness hypothesis; Corollary 1.3 with its short proof; Proposition 6.1 and Example 6.3.

2. Preliminaries

Sieves and closure operators. A sieve on c is a set S of morphisms with codomain c closed under precomposition; for $b\colon x \to c$, $b^*S = \{e : be \in S\}$ is a sieve on x. A Grothendieck topology J assigns to each c a set $J(c)$ of covering sieves: the maximal sieve is covering, b^*S is covering when S is, and S is covering when some covering R has b^*S covering for all $b \in R$ [3, III.2]. Covering sieves are closed under finite intersections. The *closure operator* of J is

$$j(S) = \{b\colon x \to c :\ b^*S \in J(x)\},$$

a sieve on c containing S; j is monotone, idempotent, preserves finite intersections, and commutes with pullback: $b^*(jS) = j(b^*S)$. A sieve S on c is covering iff $\mathrm{id}_c \in j(S)$ iff $j(S)$ is maximal. These operators are the Lawvere–Tierney topologies on $\mathrm{Presh}(S)$ [3, V.1], $\Omega(c)$ being the set of sieves on c; topologies are ordered by $J \le J'$ iff $J(c) \subseteq J'(c)$ for all c, iff $j \le j'$ pointwise.

Local isomorphisms. For a map $f\colon X \to Y$ of presheaves, $y \in Y(c)$ and $x, x' \in X(c)$ put

$$\mathrm{Im}_f(y) = \{b\colon x_0 \to c :\ y{\cdot}b \in \mathrm{im} f_{x_0}\}, \quad \mathrm{Eq}(x,x') = \{b\colon x_0 \to c :\ x{\cdot}b = x'{\cdot}b\}.$$

Both are sieves, $b^*\mathrm{Im}_f(y) = \mathrm{Im}_f(y \cdot b)$ and $b^*\mathrm{Eq}(x,x') = \mathrm{Eq}(x \cdot b, x' \cdot b)$. By [3, III.7, Cor. 6], $f \in W_J$ iff $\mathrm{Im}_f(y) \in J(c)$ for all y and $\mathrm{Eq}(x,x') \in J(c)$ whenever $f(x) = f(x')$ (f is *J-locally surjective* and *J-locally injective*). In particular an injection is in W_J iff it is J-locally surjective (J-dense), and a surjection is in W_J iff it is J-locally injective.

Words and reachability. Fix topologies J, J' with closure operators j, j' and put $c := jj'$, a monotone inflationary operator on sieves commuting with pullback. For an operator w composed of j's and j''s (a *word*; $w = jj'$ means j' first), call a sieve S on c *w-reachable* if $\mathrm{id}_c \in w(S)$, equivalently $w(S)$ is maximal; *k-reachable* means c^k-reachable.

Lemma 2.1. *Let w, w' be words and S, T sieves on c.*

(i) *$b \in w(S)$ iff b^*S is w-reachable. Hence S is w-reachable iff b^*S is w-reachable for every b into c.*
(ii) *If S is w-reachable and b^*T is w'-reachable for every $b \in S$, then T is ww'-reachable.*
(iii) *If S and T are w-reachable, so is $S \cap T$; if $S \subseteq T$ and S is w-reachable, so is T; a k-reachable sieve is $(k+1)$-reachable; a J-covering or J'-covering sieve is 1-reachable.*

Proof. (i) $b^*(wS) = w(b^*S)$, and $b \in wS$ iff $\mathrm{id} \in b^*(wS)$. (ii) By (i), $S \subseteq w'(T)$, so $w(S) \subseteq w(w'(T))$ contains the identity. (iii) Each of j, j' preserves finite intersections, is monotone and is inflationary; $j(S) \subseteq jj'(S)$ and $j'(S) \subseteq jj'(S)$. □

The join topology. Let $K = J \vee J'$ be the least topology above J and J' and j_K its closure operator. Define the transfinite iterates $c^0(S) = S$, $c^{\alpha+1}(S) = c(c^\alpha(S))$ and $c^\lambda(S) = \bigcup_{\alpha<\lambda} c^\alpha(S)$ for limit λ; they increase with α, commute with pullback, and $c^\alpha(c^\beta(S)) = c^{\beta+\alpha}(S)$.

Lemma 2.2. *$j_K(S) = \bigcup_\alpha c^\alpha(S)$; a sieve is K-covering iff it is c^α-reachable for some ordinal α. (The sieves on c form a set, so the increasing iterates $c^\alpha(S)$ are constant from some ordinal on, and the union is attained.)*

Proof. Since $j, j' \le j_K$ and j_K is monotone and idempotent, $c^\alpha(S) \subseteq j_K(S)$ for all α by induction (unions at limits). Let $T(c)$ be the set of sieves S on c with $\mathrm{id}_c \in c^\alpha(S)$ for some α. It contains $J(c)$ and $J'(c)$ by Lemma 2.1(iii), the maximal sieve, and is stable under pullback since $b^*(c^\alpha S) = c^\alpha(b^*S)$. For transitivity let $S \in T(c)$ and $b^*S' \in T$ for all $b \in S$, say $\mathrm{id} \in c^{\alpha_b}(b^*S')$; with $\beta = \sup_b \alpha_b$ we get $b \in c^\beta(S')$ for all $b \in S$, so $S \subseteq c^\beta(S')$ and $\mathrm{id} \in c^\alpha(S) \subseteq c^{\beta+\alpha}(S')$. Thus T is a topology above J, J', hence $T \supseteq K$, and $T \subseteq K$ by the first sentence. Finally $b \in j_K(S)$ iff $b^*S \in K$ iff $\mathrm{id} \in c^\alpha(b^*S) = b^*(c^\alpha S)$ for some α iff $b \in \bigcup_\alpha c^\alpha(S)$. □

3. Uniform locality and the join

Fix topologies J, J', $K = J \vee J'$, $c = jj'$, and write $V := W_J \vee W_{J'}$.

Definition 3.1. For $k \ge 0$ let U_k be the class of maps $f\colon X \to Y$ such that every image sieve $\mathrm{Im}_f(y)$ and every equaliser sieve $\mathrm{Eq}(x, x')$ with $f(x) = f(x')$ is k-reachable. Put $U = \bigcup_k U_k$.

U_0 is the class of isomorphisms (maximal image sieves mean surjectivity, maximal equaliser sieves injectivity). A map in U_k is a K-local isomorphism *uniformly*: the ordinal of Lemma 2.2 needed for its image and equaliser sieves is the finite number k, independent of the element.

Lemma 3.2. *$U_k \subseteq U_{k+1}$, $W_J \cup W_{J'} \subseteq U_1$, $U \subseteq W_K$, and U is a sameness. Hence $V \subseteq U \subseteq W_K$.*

Proof. The first three assertions are Lemma 2.1(iii) and Lemma 2.2. Identities lie in U_0. Let $h\colon X \to Y$ and $g\colon Y \to Z$.

Composition. Let $h \in U_k$, $g \in U_l$. For $z \in Z(c)$ the sieve $S = \mathrm{Im}_g(z)$ is l-reachable; for $b \in S$ write $z \cdot b = g(y_b)$; then $b^*\mathrm{Im}_{gh}(z) = \mathrm{Im}_{gh}(z \cdot b) \supseteq \mathrm{Im}_h(y_b)$ is k-reachable, so $\mathrm{Im}_{gh}(z)$ is $(l+k)$-reachable by Lemma 2.1(ii). If $gh(x) = gh(x')$, the sieve $S = \mathrm{Eq}(h(x), h(x'))$ is l-reachable; for $b \in S$, $h(x \cdot b) = h(x' \cdot b)$, so $b^*\mathrm{Eq}(x, x') = \mathrm{Eq}(x \cdot b, x' \cdot b)$ is k-reachable; hence $\mathrm{Eq}(x, x')$ is $(l+k)$-reachable.

Cancellation of g. Let $gh \in U_k$, $g \in U_k$. If $h(x) = h(x')$ then $gh(x) = gh(x')$ and $\mathrm{Eq}(x, x')$ is k-reachable. For $y \in Y(c)$ the sieve $S = \mathrm{Im}_{gh}(g(y))$ is k-reachable; for $b \in S$ write $g(y) \cdot b = gh(x_b)$, so $g(y \cdot b) = g(h(x_b))$ and $T_b = \mathrm{Eq}(y \cdot b, h(x_b))$ is k-reachable; for $e \in T_b$, $y \cdot be = h(x_b \cdot e) \in \mathrm{im}h$, so $b^*\mathrm{Im}_h(y) \supseteq T_b$. By Lemma 2.1(ii), $\mathrm{Im}_h(y)$ is $2k$-reachable.

Cancellation of h. Let $gh \in U_k$, $h \in U_k$. Then $\mathrm{Im}_g(z) \supseteq \mathrm{Im}_{gh}(z)$ is k-reachable. If $g(y) = g(y')$, the sieve $S = \mathrm{Im}_h(y) \cap \mathrm{Im}_h(y')$ is k-reachable; for $b \in S$ write $y \cdot b = h(x_b)$, $y' \cdot b = h(x'_b)$; then $gh(x_b) = g(y) \cdot b = g(y') \cdot b = gh(x'_b)$, so $\mathrm{Eq}(x_b, x'_b)$ is k-reachable, and for $e \in \mathrm{Eq}(x_b, x'_b)$, $y \cdot be = h(x_b \cdot e) = h(x'_b \cdot e) = y' \cdot be$; so $b^*\mathrm{Eq}(y, y') \supseteq \mathrm{Eq}(x_b, x'_b)$ and $\mathrm{Eq}(y, y')$ is $2k$-reachable. □

Lemma 3.3 (peeling)**.** *Let $w = j_1 j_2 \cdots j_n$ be a word, each $j_i \in \{j, j'\}$ with topology J_i.*

(a) *If $f\colon X \to Y$ is surjective and $\mathrm{Eq}(x, x')$ is w-reachable whenever $f(x) = f(x')$, then $f = f_1 \circ \cdots \circ f_n$ with f_i a surjection in W_{J_i}.*
(b) *If $f\colon X \to Y$ is injective and every $\mathrm{Im}_f(y)$ is w-reachable, then $f = f_1 \circ \cdots \circ f_n$ with f_i an injection in W_{J_i}.*

Proof. Induction on n, proving that f is a composite of n such maps followed by an isomorphism, which for $n \geq 1$ can be absorbed into the last factor; for $n = 0$ the hypothesis says that f is bijective. (a) Let R be the relation on $X(c)$: $x \,R\, x'$ iff $f(x) = f(x')$ and $\mathrm{Eq}(x, x') \in J_n(c)$. It is an equivalence relation, since covering sieves contain the maximal sieve and are closed under intersection, and a congruence, since $b^*\mathrm{Eq}(x, x') = \mathrm{Eq}(x \cdot b, x' \cdot b)$ and covering sieves pull back. The quotient $q\colon X \twoheadrightarrow X/R$ is J_n-locally injective by construction, hence in W_{J_n}, and f factors as $\bar{f} \circ q$ with $\bar{f}$ surjective. If $\bar{f}[x] = \bar{f}[x']$ then

$$\mathrm{Eq}([x], [x']) = \{b : \ x \cdot b \,R\, x' \cdot b\} = \{b : \ b^*\mathrm{Eq}(x, x') \in J_n\} = j_n(\mathrm{Eq}(x, x')),$$

and since $\mathrm{id} \in j_1 \cdots j_{n-1}(j_n \mathrm{Eq}(x, x'))$ this sieve is $j_1 \cdots j_{n-1}$-reachable. Induction applied to $\bar{f}$ gives the factorisation. (b) Regard X as a subpresheaf of Y and let $Z(c) = \{y \in Y(c) : \mathrm{Im}_f(y) \in J_n(c)\}$, the J_n-closure of X in Y; it is a subpresheaf containing X, the inclusion $X \hookrightarrow Z$ is J_n-dense, hence in W_{J_n}, and for $y \in Y(c)$,

$$\mathrm{Im}_Z(y) = \{b : \ y \cdot b \in Z\} = \{b : \ b^*\mathrm{Im}_f(y) \in J_n\} = j_n(\mathrm{Im}_f(y)),$$

which is $j_1 \cdots j_{n-1}$-reachable. Induction applied to $Z \hookrightarrow Y$ gives the factorisation. □

Proof of Theorem 1.1. Let C be the closure of $W_J \cup W_{J'}$ under composition. Then $C \subseteq V \subseteq U$ by Lemma 3.2. Let $f \in U_k$ and factor $f = m \circ e$ with $e\colon X \twoheadrightarrow \mathrm{im} f$ and $m\colon \mathrm{im} f \hookrightarrow Y$. The equaliser sieves of e are those of f, and the image sieves of m are those of f; both are c^k-reachable, and $c^k = (jj')^k$ is a word of length $2k$ ending with j'. By Lemma 3.3, e is a composite of $2k$ surjections and m a composite of $2k$ injections, alternately in $W_{J'}$ and W_J, beginning with $W_{J'}$. So $U \subseteq C$, and $C = V = U$. □

Remark 3.4. In general the join of two samenesses requires the two-out-of-three saturation of the compositional closure [5]; here no cancellation is needed. The number of steps is controlled in both directions: a composite of n maps of $W_J \cup W_{J'}$ lies in U_n by Lemma 3.2, and a map of U_k is a composite of $4k$ such maps.

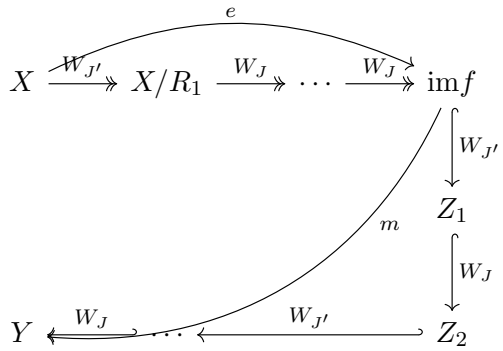


FIGURE 1. Theorem 1.1: a map in U_k is $2k$ quotients by "equal under f and locally equal" followed by $2k$ alternating closures of its image, each step a local isomorphism for one of the two topologies.

4. The criterion

Proposition 4.1. *Let S be a sieve on c, let X_S be the presheaf obtained from two copies $y(c)^{(1)}, y(c)^{(2)}$ of the representable $y(c) = S(-,c)$ by identifying $b^{(1)}$ with $b^{(2)}$ for every $b \in S$ (the pushout of $y(c) \leftarrow S \rightarrow y(c)$), and let $f_S \colon X_S \to y(c)$ be the fold map. Then $f_S \in W_K$ iff S is K-covering, and $f_S \in U_k$ iff S is k-reachable.*

Proof. The identification is compatible with restriction because S is a sieve, so X_S is a presheaf and f_S a surjective map of presheaves; its image sieves are maximal. Two distinct elements with the same image are $b^{(1)}$, $b^{(2)}$ for some $b \notin S$, and $\mathrm{Eq}(b^{(1)}, b^{(2)}) = \{e : be \in S\} = b^*S$. So $f_S \in W_K$ iff $b^*S \in K$ for all $b \notin S$; if S is maximal f_S is an isomorphism, otherwise $\mathrm{id}_c \notin S$ and the condition implies $S \in K$, while $S \in K$ implies $b^*S \in K$ for all b. Likewise $f_S \in U_k$ iff b^*S is k-reachable for all $b \notin S$ iff $b \in c^k(S)$ for all $b \notin S$ (Lemma 2.1(i)) iff $c^k(S)$ is maximal, as $c^k(S) \supseteq S$. □

Proof of Theorem 1.2. The two formulations of the condition agree: $(jj')^k \leq j_K$ always, and if every K-covering sieve is k-reachable then $b \in j_K(S)$ gives $b^*S \in K$, so $\mathrm{id} \in c^k(b^*S) = b^*c^k(S)$ and $b \in c^k(S)$; conversely $c^k = j_K$ makes every covering sieve k-reachable. If the condition holds, a map in W_K has K-covering, hence k-reachable, image and equaliser sieves, so $W_K \subseteq U_k \subseteq V$, and $V \subseteq W_K$ always. If it fails, choose for each k an object c_k and a K-covering sieve S_k on c_k that is not k-reachable, and let $F = \coprod_k f_{S_k}$. Then $F \in W_K$, since image and equaliser sieves of a coproduct of maps are those of the components (elements of a coproduct of presheaves restrict within their summand), and $F \notin U_k$ for every k because its component f_{S_k} is not in U_k (Proposition 4.1). So $F \in W_K \setminus U = W_K \setminus V$ by Theorem 1.1. □

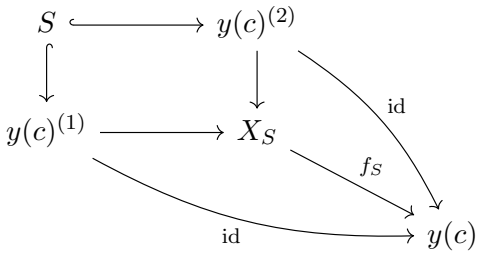


FIGURE 2. The witness of Proposition 4.1: two copies of the representable glued along the sieve S; the fold map identifies the two identities, whose equaliser sieve is S.

Remark 4.2. Lemma 2.2 and Theorem 1.2 place the question in the setting of [9, 11]: there the join $j_1 \vee j_2$ of two nuclei on a Heyting algebra or frame L is the transfinite iterate of $j_2 j_1$ [1], and the identity $(j_2 j_1)^N = j_1 \vee j_2$ for some N, the finite-iteration axiom of the lax logic of the join in [9], is exactly the condition under which the join of the two reflections of L-fuzzy sets is generated at the level of inverted maps [11, Cor. 4.3]. Theorem 1.2 is the same statement for the internal frame Ω of $\mathrm{Presh}(S)$ with the Lawvere–Tierney topologies j, j': the join of two subtopos classes is the class of the meet of the subtoposes iff the alternation of the two local operators reaches the join uniformly in finitely many steps. The condition is uniform in the sieve; it should be distinguished from the pointwise condition that the iteration terminates at each element, which suffices in $\mathrm{Same}(L)$ [11] and which, for the relativized necessities $\Box_C, \Box_{C'}$ on the frame of upsets of a poset, is the absence of infinite alternating chains characterising $\Box_C \vee \Box_{C'} = \Box_{C \cap C'}$ [10]; a disjoint union of finite alternating chains of unbounded length satisfies the pointwise condition but not the uniform one.

5. Topologies of ideal type

For an ideal A the sets $A(-, c)$ are sieves and $A(-, c) \cdot b \subseteq A(-, x)$ for $b \colon x \to c$.

Lemma 5.1. *Let D, E be idempotent ideals.*

(i) *J_D is a topology, $D(-, c)$ is its least covering sieve on c, and $J_D \leq J_E$ iff $D \supseteq E$.*

(ii) *The closure operator of J_D is $j_D(S) = \{b : \ b \cdot D \subseteq S\}$, where $b \cdot D = \{b \circ d : d \in D(-, \mathrm{dom}\, b)\}$; hence $(j_D j_E)^k(S) = \{b : \ b \cdot (DE)^k \subseteq S\}$, and S is k-reachable for J_D, J_E iff $S \supseteq (DE)^k(-, c)$. A map is in U_k iff it is* locally bijective along $(DE)^k$*: $f(x) = f(x')$ implies $x \cdot a = x' \cdot a$ for all $a \in (DE)^k(-, c)$, and $y \cdot a \in \mathrm{im} f$ for all $y \in Y(c)$ and $a \in (DE)^k(-, c)$.*

(iii) *For every k and c the sieve $(DE)^k(-, c)$ is $(J_D \vee J_E)$-covering.*

(iv) *If* $(DE)^m = (DE)^{m+1} =: I'$ *then* I' *is idempotent,* $I' \subseteq D \cap E$, $DI' = I'D = EI' = I'E = I'$, $(ED)^{m+1} = I'$, *every alternating product of at least* $2(m+1)$ *factors* D, E, *in either order, equals* I', *and* $J_D \vee J_E = J_{I'}$. *(The chain* $(ED)^k$ *need not have stabilised at index* m*: Example 5.3.)*

Proof. (i) is [6, Thm. 5.1]: the maximal sieve contains $D(-,c)$; pullback of a sieve containing $D(-,c)$ along $b\colon x \to c$ contains $D(-,x)$ because D is a left ideal; and if $R \supseteq D(-,c)$ and each a^*R' with $a \in R$ contains $D(-,\mathrm{dom}\, a)$, then for $d \in D(-,c)$, $d = d_1 d_2$ with $d_i \in D$ by idempotency, $d_1 \in R$ and $d_2 \in d_1^* R'$, so $d \in R'$. (ii) $b \in j_D(S)$ iff $b^*S \supseteq D(-,\mathrm{dom}\, b)$ iff $b \cdot D \subseteq S$; then $b \in j_D j_E(S)$ iff $(b \circ d) \cdot E \subseteq S$ for all d iff $b \cdot DE \subseteq S$, and inductively for $(DE)^k$; $\mathrm{id}_c \cdot (DE)^k = (DE)^k(-,c)$. For a map f, $\mathrm{Im}_f(y) \supseteq (DE)^k(-,c)$ says $y \cdot a \in \mathrm{im} f$ for $a \in (DE)^k(-,c)$, and $\mathrm{Eq}(x,x') \supseteq (DE)^k(-,c)$ says $x \cdot a = x' \cdot a$. (iii) is Lemma 2.2: $(DE)^k(-,c)$ is k-reachable. (iv) $I'I' = (DE)^{2m} = I'$; $I' \subseteq DE \subseteq D \cap E$; $DI' = DD\,E(DE)^{m-1} = (DE)^m = I'$ as $DD = D$, and similarly $I'E = I'$; $I'D \supseteq I'I' = I'$ as $I' \subseteq D$, and $I'D \subseteq I'$ as I' is a right ideal, so $I'D = I'$, and likewise $EI' = I'$; then $(ED)^{m+1} = E(DE)^m D = EI'D = I'$. An alternating product of at least $2(m+1)$ factors contains $(DE)^{m+1}$ or $(ED)^{m+1}$ as a block of consecutive factors; the block equals I', and multiplying I' by D or E on either side returns I'. For the join, $J_{I'} \geq J_D, J_E$ since $I' \subseteq D, E$, and any topology above J_D and J_E has $I'(-,c) = (DE)^m(-,c)$ covering by (iii), hence is above $J_{I'}$. □

Proof of Corollary 1.3. By Lemma 5.1(ii) and Theorem 1.1, $W_{J_D} \vee W_{J_E}$ is the class of maps locally bijective along some $(DE)^k$. If $(DE)^m = (DE)^{m+1} = I'$, then $K = J_{I'}$ by Lemma 5.1(iv), a K-covering sieve contains $I'(-,c) = (DE)^m(-,c)$ and is therefore m-reachable; Theorem 1.2 gives $V = W_K = W_{J_{I'}}$, and Theorem 1.1 gives the factorisation into $4m$ maps, since $W_{J_{I'}} = U_m$. If the chain does not stabilise, then $(DE)^k \neq (DE)^{k+1}$ for every k (once two consecutive terms agree, all later ones do), so there are c_k with $(DE)^k(-,c_k) \not\subseteq (DE)^{k+1}(-,c_k)$; the sieve $S_k = (DE)^{k+1}(-,c_k)$ is K-covering by Lemma 5.1(iii) and not k-reachable by Lemma 5.1(ii), so Theorem 1.2 gives $V \neq W_K$. The witnesses of Proposition 4.1 are here two copies of $y(c_k)$ glued along $(DE)^{k+1}(-,c_k)$, whose fold map is locally bijective along $(DE)^{k+1}$ but identifies the two identities, which restrict along a morphism of $(DE)^k \setminus (DE)^{k+1}$ into c_k to distinct elements. □

Remark 5.2. The first version of this paper proved the stabilising direction by the route of [8]: $2r$ alternating separated quotients, $r = m+1$, followed by $4r$ alternating plus constructions, with a retraction of the representing bimodule $I' \otimes I'$ onto the alternating tensor power. Theorem 1.1 replaces this by the single observation $W_{J_{I'}} = U_m \subseteq V$ and improves the bound to $4m$ maps. The plus tower is nevertheless of independent interest, as it need not stabilise even when the chain does [8, §7].

Example 5.3. The two chains $(DE)^k$ and $(ED)^k$ need not stabilise at the same index. Let S be the category with two objects $0, 1$ and one non-identity arrow $a\colon 0 \to 1$, and let $D = \{\mathrm{id}_0, a\}$, $E = \{\mathrm{id}_1, a\}$ be the ideals of arrows factoring through 0, respectively through 1; both are idempotent. Then $DE = \emptyset$ (no arrow of E has codomain 0), while $ED = \{a\}$. So $(DE)^1 = (DE)^2 = \emptyset = I'$, $m = 1$, but $(ED)^1 = \{a\} \neq I'$, and only $(ED)^2 = \emptyset = I'$. The join $J_D \vee J_E = J_\emptyset$ is the degenerate topology, as the empty sieve on 0 is J_E-covering and the sieve $\{a\}$ on 1 is J_D-covering.

Example 5.4. On a finite category every chain of ideals stabilises, so Corollary 1.3 contains [8, Thm. 1.1]. On $S = (\mathbb{N}, \leq)$ with D and E the ideals of arrows $n \to m$ passing through an even, respectively an odd, point, $(DE)^k$ consists of the arrows $n \to m$ admitting k alternating even and odd points between n and m, a strictly decreasing chain with empty intersection; the join topology is degenerate and W_K is everything, while V is not [7, Thm. 7.1]. Example 6.3 describes V.

Example 5.5. For a monoid M, viewed as a category with one object, ideals are the two-sided ideals of M and the criterion is the stabilisation of the alternating products $DE \supseteq DEDE \supseteq \cdots$ in the quantale of ideals of M. Every finite monoid satisfies it, with m at most the number of elements. For infinite monoids the chain can decrease forever, and then the join of the two classes of local isomorphisms of M-sets is the class of maps locally bijective along some $(DE)^k$, strictly smaller than the class for the join topology.

6. Ideals generated by objects

For a set A of objects of S let D_A be the set of morphisms factoring through an object of A. It is an ideal, idempotent since $f = g \circ h$ with $h\colon x \to a$, $g\colon a \to y$ has $g, h \in D_A$; on a finite category every idempotent ideal is of this form after Cauchy completion [8, §2], but not in general: on $(\mathbb{Q}, \leq)$ the ideal of strict inequalities is idempotent and contains no identity. Write $W_A := W_{J_{D_A}}$.

Proposition 6.1. *Let A, B be sets of objects of S.*

(a) *W_A is the class of maps f with f_a bijective for every $a \in A$; $(D_A D_B)^k$ is the set of morphisms factoring through a chain $b_1 \to a_1 \to b_2 \to \cdots \to b_k \to a_k$ with $b_i \in B$, $a_i \in A$, and contains $D_{A\cap B}$. The join $W_A \vee W_B$ is the class of maps f such that, for some k, f is bijective along every such chain: if $f(x) = f(x')$ then $x \cdot g = x' \cdot g$, and every $y \cdot g$ lies in the image, for every morphism g factoring through a chain of length k.*

(b) *$W_A \vee W_B = W_{A\cap B}$ if and only if for some N every morphism factoring through such a chain of length N factors through an object of $A \cap B$, i.e. $(D_A D_B)^N \subseteq D_{A\cap B}$.*

(c) *If $S = P$ is a poset whose intervals are finite, the condition in (b) holds if and only if the strictly alternating chains $b_1 < a_1 < \cdots <$*

$b_k < a_k$ ($b_i \in B$, $a_i \in A$) contained in intervals disjoint from $A \cap B$ have a bounded number of elements; if they do not, the chain $(D_A D_B)^k$ does not stabilise and $W_A \vee W_B \subsetneq W_{J_{D_A} \vee J_{D_B}}$.

Proof. (a) If f_a is bijective for $a \in A$, then for $x, x' \in X(c)$ with $f(x) = f(x')$ and $g = g_1 g_2 \in D_A(-, c)$ with $g_1 \colon a \to c$, $a \in A$, we get $f_a(x \cdot g_1) = f_a(x' \cdot g_1)$, so $x \cdot g_1 = x' \cdot g_1$ and $x \cdot g = x' \cdot g$; and for $y \in Y(c)$, $y \cdot g_1 = f_a(z)$ for some z, so $y \cdot g = f(z \cdot g_2)$. Thus f is locally bijective along D_A, and $f \in W_A$ by Lemma 5.1(ii). Conversely $\mathrm{id}_a \in D_A(-, a)$ for $a \in A$, so local bijectivity along D_A gives f_a bijective. A morphism of $D_A D_B$ is $d \circ e$ with e factoring through some $b \in B$ and d through some $a \in A$, i.e. a morphism factoring through a chain $b \to a$; iterating gives the description of $(D_A D_B)^k$, and a morphism through $c \in A \cap B$ factors through the chain $c \to c \to \cdots \to c$ of identities. The description of the join is Corollary 1.3. (b) If $(D_A D_B)^N \subseteq D_{A \cap B}$ then $(D_A D_B)^N \subseteq D_{A \cap B} \subseteq (D_A D_B)^{N+1} \subseteq (D_A D_B)^N$, so the chain stabilises at $I' = D_{A \cap B}$ and Corollary 1.3 gives $W_A \vee W_B = W_{J_{I'}} = W_{A \cap B}$. Conversely, $D_{A \cap B} \subseteq D_A \cap D_B$ gives $J_{D_{A \cap B}} \geq K := J_{D_A} \vee J_{D_B}$ by Lemma 5.1(i), hence $W_{A \cap B} \supseteq W_K \supseteq W_A \vee W_B$. If $W_A \vee W_B = W_{A \cap B}$, then $W_A \vee W_B = W_K$, the chain stabilises at some $I' = (D_A D_B)^m$ by Corollary 1.3, and $W_{J_{I'}} = W_K = W_{J_{D_{A \cap B}}}$. A topology is determined by the class of maps inverted by its sheafification, since the sheaves are the objects local with respect to that class (a reflective subcategory is recovered from the maps its reflector inverts) and subtoposes of $\mathrm{Presh}(S)$ correspond bijectively to topologies [3, Chap. V]; so $J_{I'} = J_{D_{A \cap B}}$, $I' = D_{A \cap B}$ by Lemma 5.1(i), and $N = m$ works. (c) In a poset, $(n \leq m) \in (D_A D_B)^k$ iff there are $n \leq b_1 \leq a_1 \leq \cdots \leq b_k \leq a_k \leq m$ with $b_i \in B$, $a_i \in A$; if the interval $[n, m]$ is disjoint from $A \cap B$, an equality $b_i = a_i$ or $a_i = b_{i+1}$ would produce a point of $A \cap B$, so the chain is strict. Hence $(D_A D_B)^k \setminus D_{A \cap B}$ consists of the pairs $n \leq m$ with $[n, m]$ disjoint from $A \cap B$ and containing a strictly alternating chain with $2k$ elements. If these chains have at most L elements, then $(D_A D_B)^k = D_{A \cap B}$ for $2k > L$. If their sizes are unbounded, $(D_A D_B)^k \neq D_{A \cap B}$ for every k, while $\bigcap_k (D_A D_B)^k = D_{A \cap B}$ because a chain with more elements than the finite interval $[n, m]$ is not strict; a descending chain whose members all differ from its intersection does not stabilise, and Corollary 1.3 gives the strict inclusion. □

Example 6.2. On $P = (\mathbb{N}, \leq)$ let A be the even numbers and B the odd numbers together with the multiples of 4, so $A \cap B = 4\mathbb{N}$. An interval disjoint from $4\mathbb{N}$ lies in some $[4j+1, 4j+3]$, whose longest strictly alternating chain is $4j+1 < 4j+2 < 4j+3$; so $L = 3$, $(D_A D_B)^2 = D_{4\mathbb{N}}$, and a map of presheaves on $\mathbb{N}$ lies in the join of "bijective at the even points" and "bijective at the odd points and at the multiples of 4" if and only if it is bijective at the multiples of 4.

Example 6.3. On $P = (\mathbb{N}, \leq)$ let A be the even and B the odd numbers. A presheaf is a tower $\cdots \to X(2) \to X(1) \to X(0)$ of sets, and $(D_A D_B)^k(-, c)$

consists of the $n \leq c$ with $c - n \geq 2k - 1$, or $\geq 2k$ when n is even. By Proposition 6.1(a), $W_A \vee W_B$ is the class of maps of towers $f\colon X \to Y$ that are *bijective up to a shift*: for some s, $f_c(x) = f_c(x')$ implies $x|_{c-s} = x'|_{c-s}$, and every $y \in Y(c)$ has $y|_{c-s} \in \mathrm{im} f_{c-s}$ (for $c \geq s$; for $c < s$ the conditions are void). Such a map becomes an isomorphism of pro-objects: it has the inverse up to shift $g_n\colon Y(n+2s) \to X(n)$, $y \mapsto x|_n$ for any $x \in X(n+s)$ with $f(x) = y|_{n+s}$, which is well defined and compatible with restriction. The converse fails: a pro-isomorphism of towers only needs, for each n, *some* later stage $m(n)$ at which the inverse is defined, and the join requires the uniform bound $m(n) = n + 2s$. For instance let $X(n) = \{0, 1\}$ for all n, with the restriction $X(n+1) \to X(n)$ constant with value 0 when n is a power of 2 and the identity otherwise, and let $f\colon X \to 1$ be the map to the terminal tower. The section $1 \to X$ picking 0 is a pro-inverse, since for every n the composite restriction from some later stage to n is constant; but on an interval $[n, n+s]$ containing no power of 2 the two elements of $X(n+s)$ are identified by f and remain distinct at n, so f is bijective up to no uniform shift and $f \notin W_A \vee W_B$ (although $f \in W_K$, as W_K is the class of all maps). So in Same(Presh($\mathbb{N}$)) the join of "bijective at the even points" and "bijective at the odd points" is a uniform version of pro-isomorphism, strictly between the two classes and the pro-isomorphisms. The obstruction $\lim_{\mathbb{N}^{\mathrm{op}}}$ used in [7, Thm. 7.1] inverts every pro-isomorphism, and more (the projection $\{m \geq n\} \times \{0, 1\} \to \{m \geq n\}$ has empty limits on both sides and is no pro-isomorphism), so it separates the join from W_K without describing it: $W_A \vee W_B \subsetneq W_{\text{pro-iso}} \subsetneq W_{\lim} \subseteq W_K$.

Remark 6.4. For a set A of objects, the restriction functor along the full inclusion $i\colon A \hookrightarrow S$ preserves limits and has a fully faithful right adjoint, the right Kan extension Ran_i (and a left adjoint, the left Kan extension). Its essential image is a reflective subcategory whose reflector inverts exactly W_A, hence it is the subcategory of J_{D_A}-sheaves: $\mathrm{Sh}(S, J_{D_A}) \simeq \mathrm{Presh}(A)$, an essential subtopos of Presh(S) in the sense of [2]. The meet of the subtoposes Presh(A) and Presh(B) is the subtopos of $(J_{D_A} \vee J_{D_B})$-sheaves. Proposition 6.1(b) says that $W_A \vee W_B = W_{A \cap B}$ iff this meet is equivalent to Presh($A \cap B$) *and* is reached by a bounded alternation; the first condition alone does not suffice: for the even and the odd numbers on $\mathbb{N}$ the meet is equivalent to the degenerate subtopos Presh($\emptyset$), yet the join of the samenesses is not $W_\emptyset$ but the class of maps bijective up to a uniform shift.

7. Remarks and questions

Remark 7.1. The proofs of Section 3 use only the closure operators of the two topologies on subobjects, image factorisations, quotients by equivalence relations and the description of local isomorphisms by local injectivity and surjectivity. In an arbitrary topos with Lawvere–Tierney topologies j, j' one can define U_k by requiring that Y be the $2k$-fold alternating closure of $\mathrm{im} f$ and that the kernel pair of f be contained in the $2k$-fold alternating closure

of the diagonal of X; the peeling argument then goes through with closures of subobjects, and we expect Theorems 1.1 and 1.2 to hold in this generality, with Lemma 3.2 proved in the internal language. We have not carried this out.

Question 7.2. Theorem 1.2 reduces join preservation to the identity $(jj')^k = j \vee j'$ on Ω. Which pairs of topologies on a given infinite category satisfy it? For ideal topologies this is the chain condition of Corollary 1.3 and for object-generated ideals the zigzag condition of Proposition 6.1; a structural description for topologies that are not of ideal type, for instance in terms of the two subtoposes, is open.

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